Unnamed_Unit

Unit 4 Code Quest Code Variation for 4-Level Mood Lamp

PDF Documents

PDF Document 1: 17318448999891-Chapter 6_ unit4 Code Quest.pdf

This PDF document is included in the unit materials.

UNIT 4 CODE QUEST

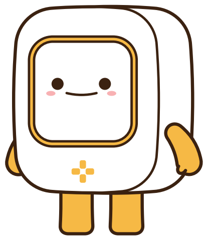
Applying Knowledge (code & debuging)

CODE

```
network
repeat forever
    if button 1 is clicked then
        variable 1 addition
        wait 0.5 seconds
    if mode = 1 then
        set LED 1 light to (red)        RED FIRE
    if mode = 2 then
        set LED 1 light to (orange)     ORANGE FIRE
    if mode = 3 then
        set LED 1 light to (blue)       BLUE FIRE
    if mode = 4 then
        turn off LED 1
```

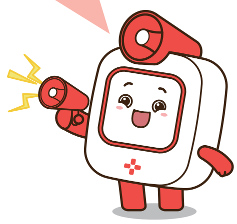
TIPS

Connecting Concepts: As we've learned about the relationship between temperature and the color of fire, let's use this knowledge to set up our mood lamp. This helps us see how science can be applied in real-world projects!



THINK ABOUT IT!

Even though our code might look a bit long, it's not as hard as it seems! Let's dive into the core and see how it works.



DEBUGGING

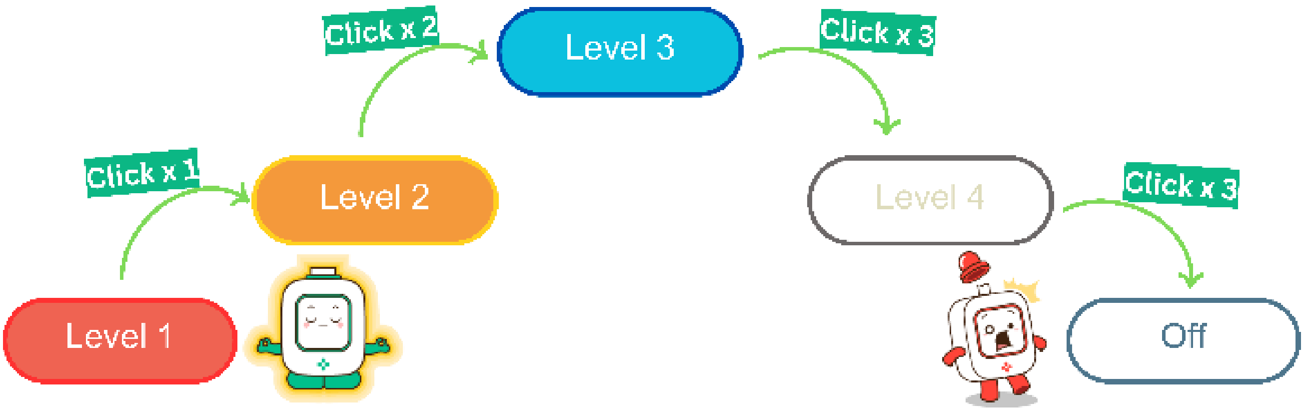
1 **What if** the wait block is missing in our code? What might happen to the way our mood lamp changes colors?

2 **What if** we use a toggle on switch instead of a clicked button to change the colors? How would the lamp's behavior change?

Missions

Misson1

1 **Flowchart:** Let' s draw a flowchart where the LED is to turn off after showing white color and then restarting. This change will transform our mood lamp into a 4-brightness lamp!



YOUR FLOWCHART

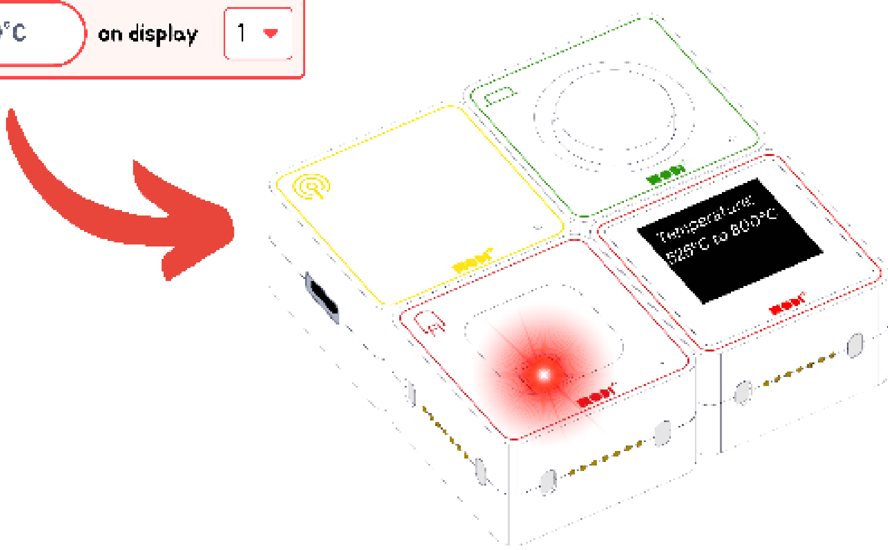
2 **Based on the flowchart, program your device!**

Misson2

1 **Let's use the Display module to show the temperature information for each LED color.**
First, fill in the blanks with the temperature information to display.

Flame Color	Temperature
RED	ex) 525°C to 800°C
ORANGE	
BLUE	
WHITE	

2 **Now, program your mood lamp using the temperature information we just filled in!**
Let's see how well we can match the LED colors with their corresponding temperatures.



TIPS

The blocks shown in the image contain everything you need for this mission. Think carefully about where each block fits into your code to make your mood lamp work as planned!